National and Kapodistrian University of Athens (NKUA)
MSc in Data Science and Information Technologies (DSIT)
Course: Algorithms in Structural Bioinformatics *(2025–2026)*

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1a)

The 3D structure of ancient VanA, PDB ID **3SE7**, was determined by **X-ray diffraction / X-ray crystallography**. The reported resolution is **3.07 Å**.

1b)

The PDB entry **3SE7** contains **six protein chains**, labeled **A, B, C, D, E, and F**. Each chain corresponds to **VanA** and has **346 residues**.

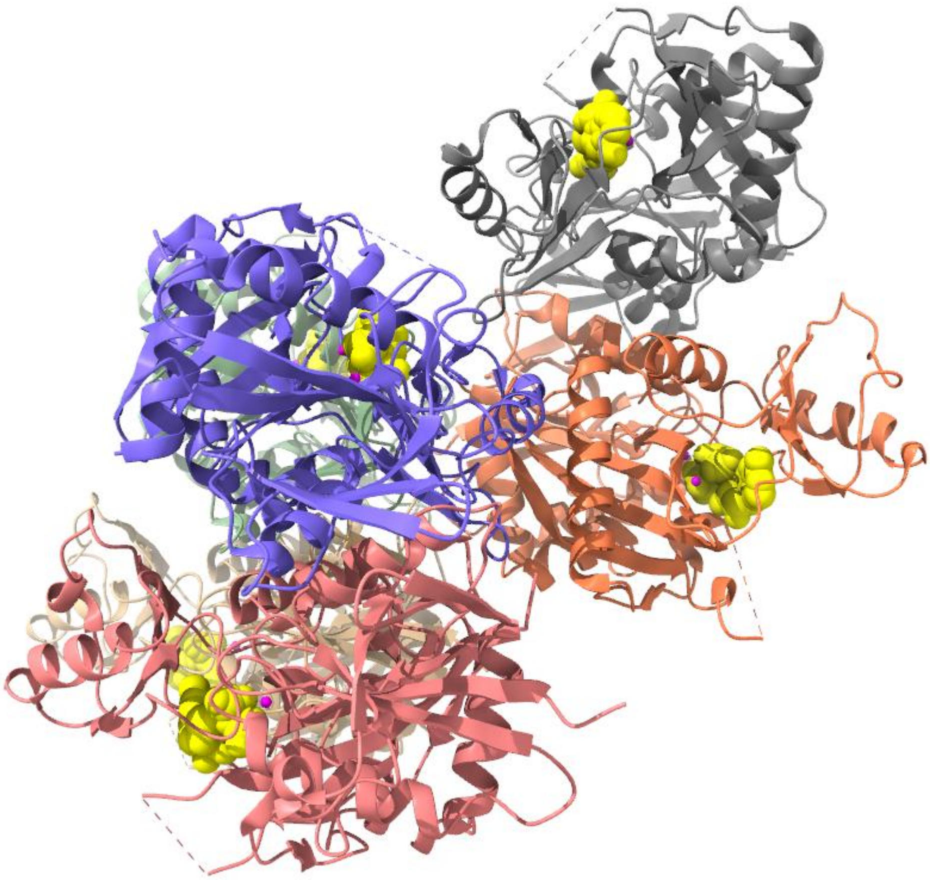
1c)

The structure **3SE7** contains **235 water molecules**.

1d)

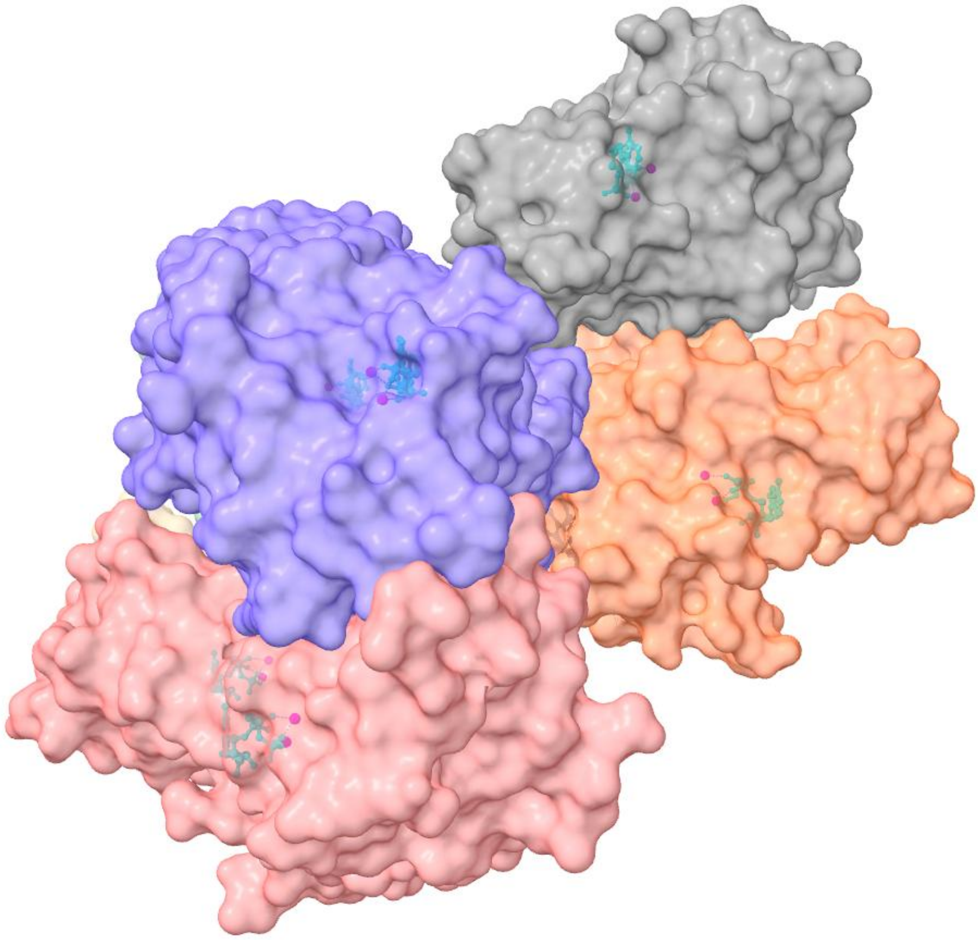
The ligands present in the structure **3SE7** are **ATP (adenosine-5′-triphosphate)** and **MG (magnesium ion, Mg^{2+})**.

2a)



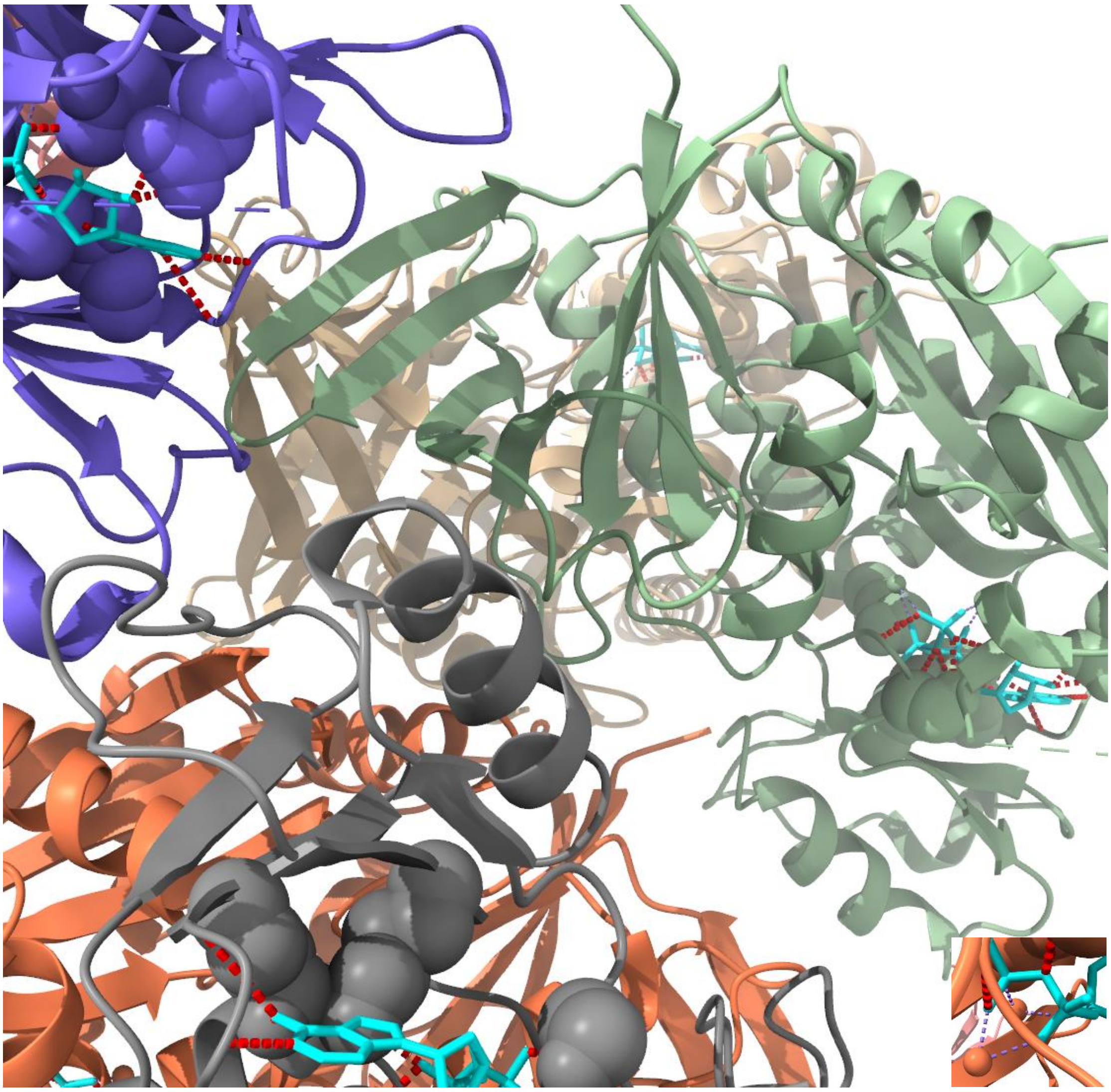
The overall structure of ancient VanA, PDB ID **3SE7**, was visualized in **UCSF ChimeraX**. The protein is shown in **cartoon representation**, with α-helices, β-strands and loops visible. The six protein chains **A, B, C, D, E and F** are displayed in different colours. The ligands are shown explicitly: **ATP** molecules are displayed as **yellow spheres**, while **Mg^{2+} ions** are displayed as **pink spheres**.

2b)



The molecular surface representation of ancient VanA, PDB ID **3SE7**, was generated using **UCSF ChimeraX**. The protein chains are shown as **semi-transparent molecular surfaces** in different colours. The ligands are shown explicitly in **ball-and-stick/sphere representation**: **ATP** molecules are displayed in **cyan**, while **Mg^{2+} ions** are displayed as **magenta spheres**.

2c)



The ligand–protein interactions of ancient VanA, PDB ID **3SE7**, were visualized using **UCSF ChimeraX**. The protein is shown in **cartoon representation**, while the ligands are displayed in **cyan**. The hydrogen bond interactions between the ligands, mainly **ATP**, and nearby protein residues are shown as **red dashed lines**. In addition, the **Mg^{2+} ions** are shown as spheres, and their **metal-coordination interactions** are displayed as **purple dashed lines** (see inset - these purple dashed lines represent coordination of Mg^{2+} mainly with oxygen atoms from the ATP phosphate groups and/or nearby coordinating atoms, and should not be interpreted as hydrogen bonds).

3a)

matchmaker #2/A,B to #1/A,B

Parameters	
Chain pairing	bb
Alignment algorithm	Needleman-Wunsch
Similarity matrix	BLOSUM-62
SS fraction	0.3
Gap open (HH/SS/other)	18/18/6
Gap extend	1
SS matrix	**H S O** **H** 6 -9 -6 **S** 6 -6 **O** 4
Iteration cutoff	2

Matchmaker 3se7, chain A (#1) with 1e4e, chain A (#2), sequence alignment score = 1226.6
RMSD between 303 pruned atom pairs is 0.740 angstroms; (across all 327 pairs: 1.815)

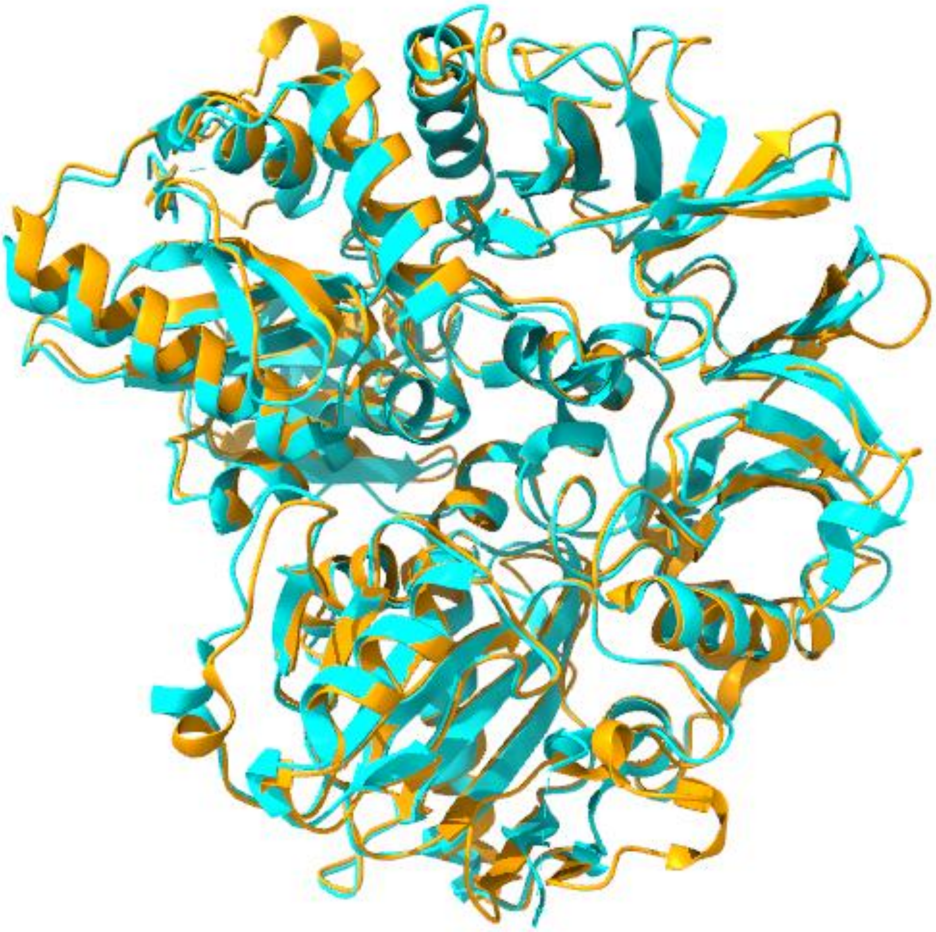
matchmaker #2/B to #1/B

Parameters	
Chain pairing	bb
Alignment algorithm	Needleman-Wunsch
Similarity matrix	BLOSUM-62
SS fraction	0.3
Gap open (HH/SS/other)	18/18/6
Gap extend	1
SS matrix	**H S O** **H** 6 -9 -6 **S** 6 -6 **O** 4
Iteration cutoff	2

Matchmaker 3se7, chain B (#1) with 1e4e, chain B (#2), sequence alignment score = 1189.7
RMSD between 301 pruned atom pairs is 0.695 angstroms; (across all 327 pairs: 1.992)

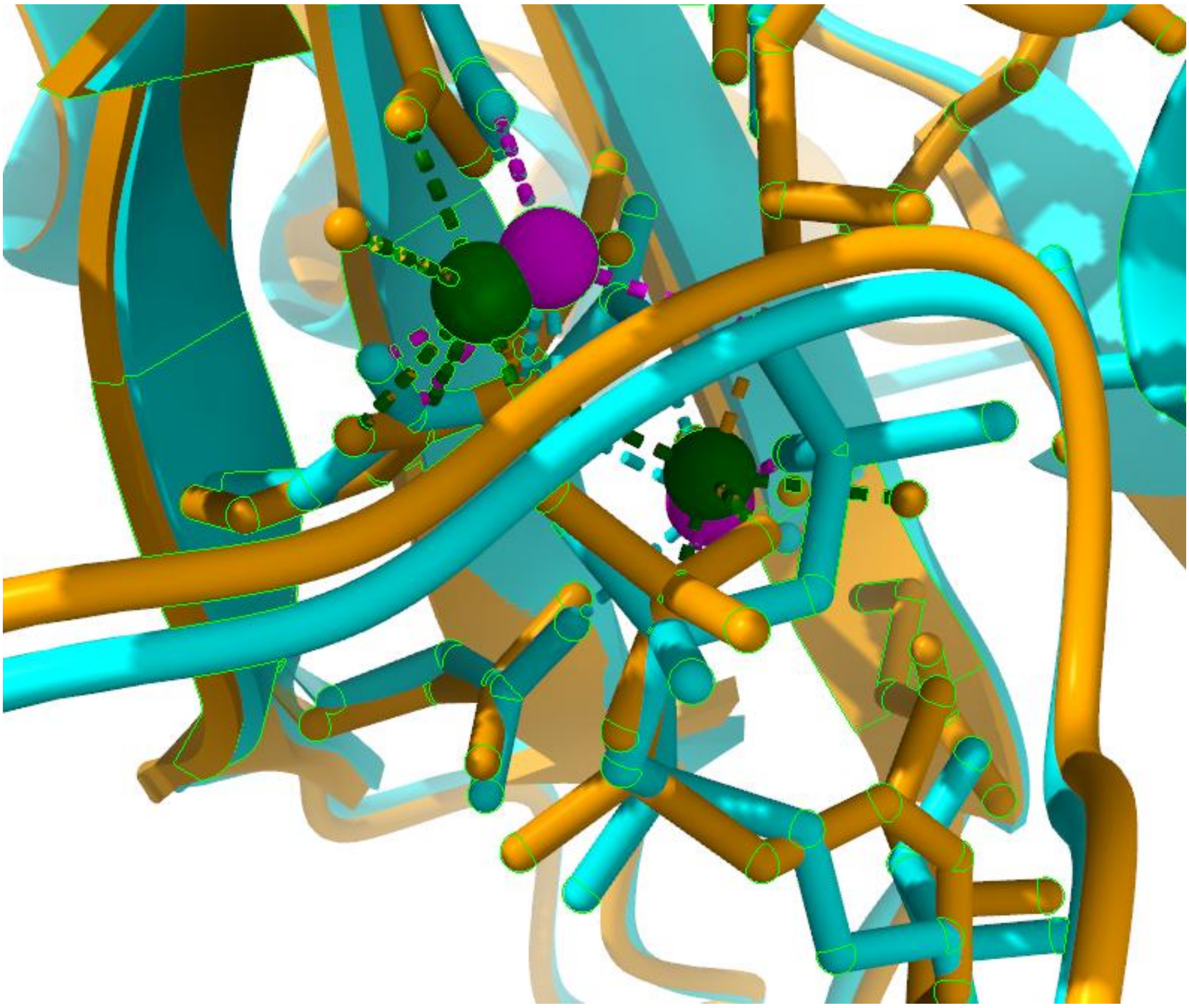
The structures **3SE7** and **1E4E** were structurally aligned in **UCSF ChimeraX** using the **Matchmaker** tool. Because **1E4E** contains two protein chains (**A** and **B**) and **3SE7** contains corresponding VanA chains, the comparison was performed separately for equivalent chains. Chain **A** of **1E4E** was aligned to chain **A** of **3SE7**, giving an RMSD of **0.740 Å over 303 pruned atom pairs**; across all **327 atom pairs**, the RMSD was **1.815 Å**. Chain **B** of **1E4E** was aligned to chain **B** of **3SE7**, giving an RMSD of **0.695 Å over 301 pruned atom pairs**; across all **327 atom pairs**, the RMSD was **1.992 Å**. These low pruned RMSD values indicate strong structural similarity between the corresponding VanA chains in the two structures.

3b)



The two structures, **3SE7** and **1E4E**, were superposed using the **Matchmaker** tool in **UCSF ChimeraX**. Since **1E4E** contains two protein chains, chains **A** and **B** of **1E4E** were aligned separately onto the corresponding chains **A** and **B** of **3SE7**. The superposed structures are shown in **cartoon representation**, with **3SE7 chains A–B** coloured **cyan** and **1E4E chains A–B** coloured **orange**. The close overlap of the two structures is consistent with the low RMSD values obtained from Matchmaker: **0.740 Å** for chain A and **0.695 Å** for chain B.

3c)



The Mg^{2+} binding site of the superposed **3SE7** and **1E4E** structures was visualized using **UCSF ChimeraX**. Chains **A** and **B** of **1E4E** were superposed onto the corresponding chains **A** and **B** of **3SE7** using the **Matchmaker** tool. The protein structures are shown in cartoon representation, with **3SE7** coloured **cyan** and **1E4E** coloured **orange**. The view is focused on the metal-binding site. **Mg^{2+} ions from 3SE7** are shown in **magenta**, whereas **Mg^{2+} ions from 1E4E** are shown in **green**. Nearby ligands and coordinating protein residues are shown as sticks, and dashed lines indicate metal-coordination interactions. This close-up highlights local differences in the Mg^{2+} binding environment between the two superposed structures.